

INTRODUCTION TO APPLIED NUMERICAL METHODS

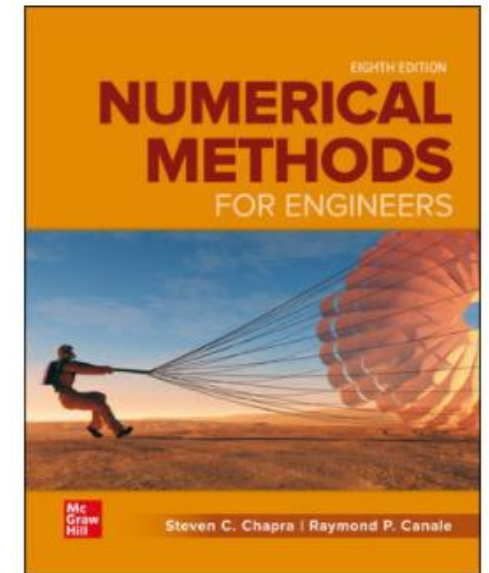
Significant Figures and Round-off Error

Lang Yuan, Associate Professor

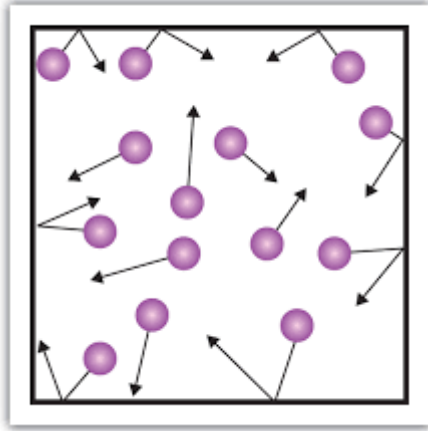


ABOUT MYSELF AND THE COURSE

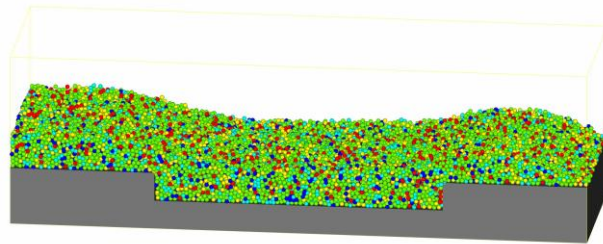
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 - Virtual: MWF 1:00PM – 2:00PM (Through Microsoft Teams)
 - In person: through email appointment
 - Feel free to contact me anytime via Email
- Grader: Tianyu(Peter) Zhang, tz5@email.sc.edu
- Syllabus key info:
 - Textbook : Numerical Methods for Engineers by S C Chapre & R P Canale
 - *Attendance policy*:
 - Randomly recorded
 - For any reasonable absence, effective communication is high recommended
 - Grading: attendance (5%) + 7 assignments (35%) + 2 midterm exams (30%) + 1 final exam(30%)



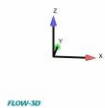
WHY NUMERICAL METHODS?



Time = 0.000



3



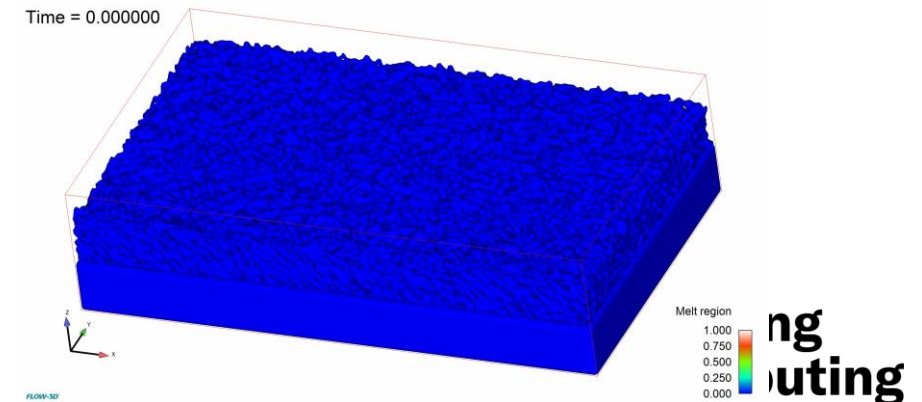
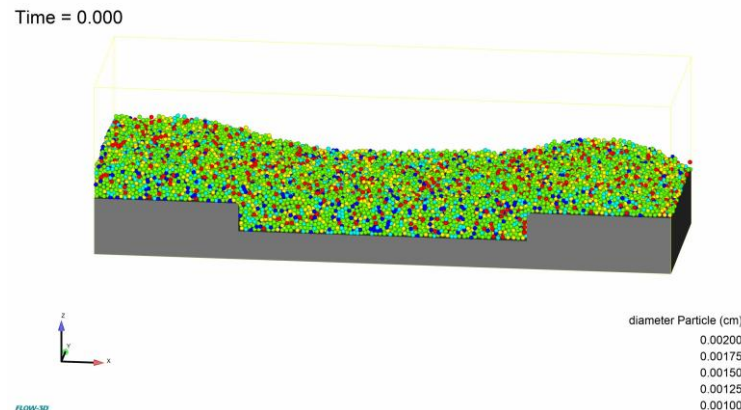
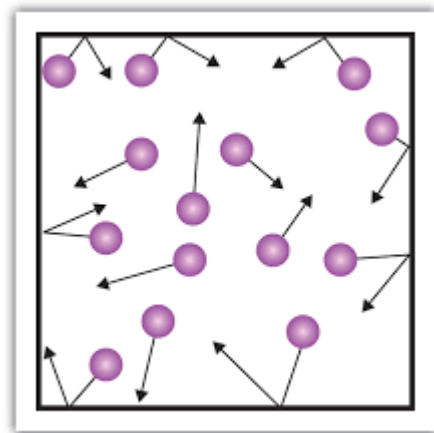
FLOW-3D

diameter Particle (cm)



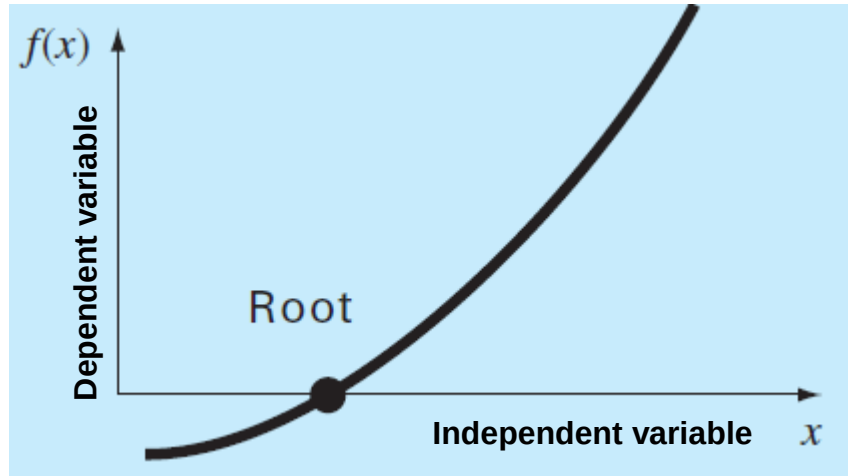
WHY NUMERICAL METHODS?

- Digitalization of the Physics Laws
 - Solved with arithmetic operations
 - Handling large mathematical problems with less approximation
 - General to problems under broader categories
 - Able to tackle both “standard” and “customized” problems
 - Excellent for learning computer languages and programming
 - **Facilitating the understanding of the underlying physics and mathematics**



SUBJECT AREAS OF THIS COURSE

Roots of equations



Math Description

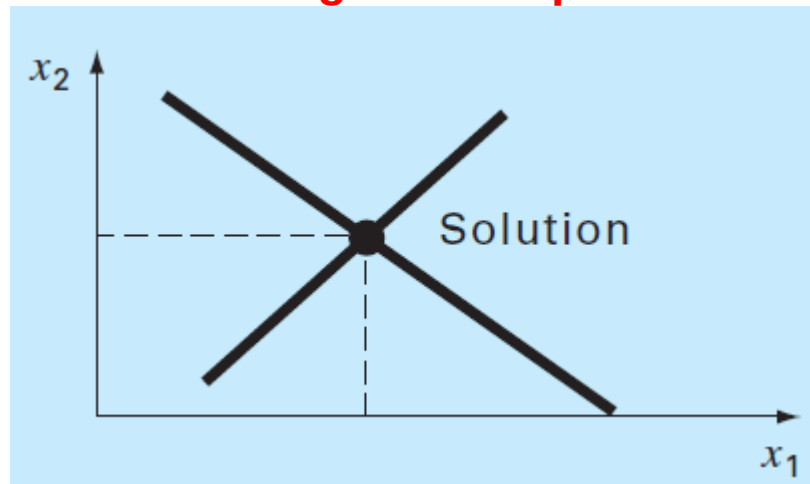
Solve: $f(x) = 0$ for x

- Determine the value of a variable or a parameter that satisfies a single nonlinear equation.

Application Examples

- Solve complex design equations for variables or parameters, e.g. diffusion equation (T)

Linear Algebraic Equations



- Given the a 's and the c 's, solve

$$a_{11}x_1 + a_{12}x_2 = c_1$$

$$a_{21}x_1 + a_{22}x_2 = c_2$$

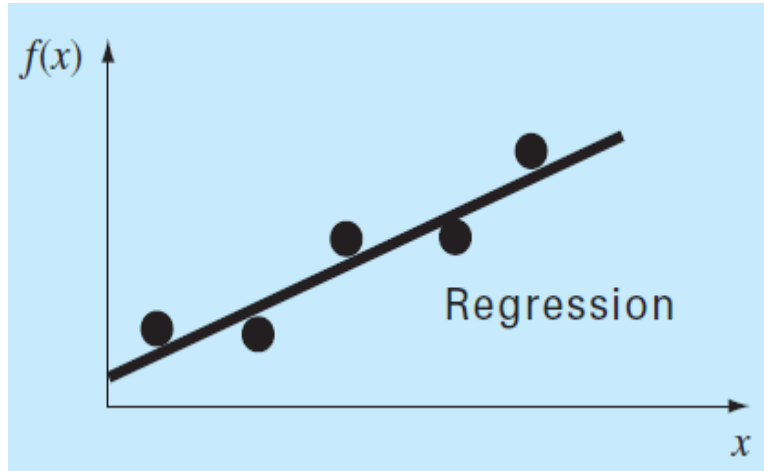
for the x 's.

Determine values that simultaneously satisfy a set of linear equation.

- Modeling of large systems of interconnected elements such as structures, electric circuits, and fluid networks.
- Groundwork of numerical methods (curve fitting and differential equations).

SUBJECT AREAS OF THIS COURSE

Regression



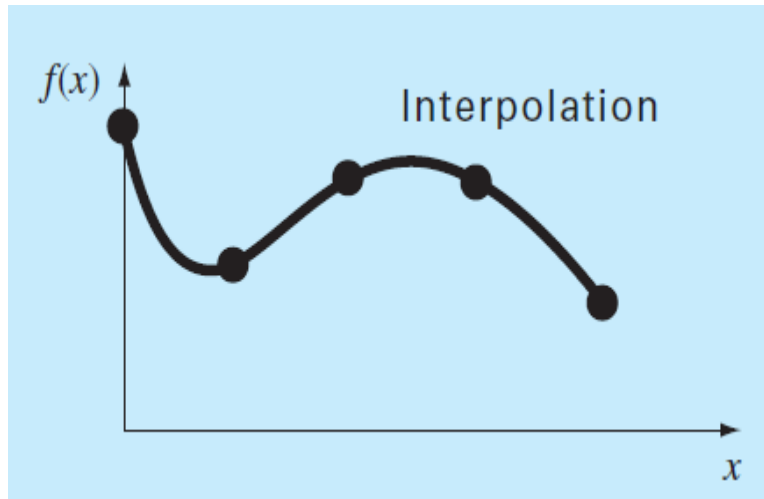
Math Description

- Fit curves to data points
- A significant degree of error associated with the data
- Representing **the general trend** of the data without necessarily matching any individual points

Application Examples

- Fitting experimental Data

Interpolation

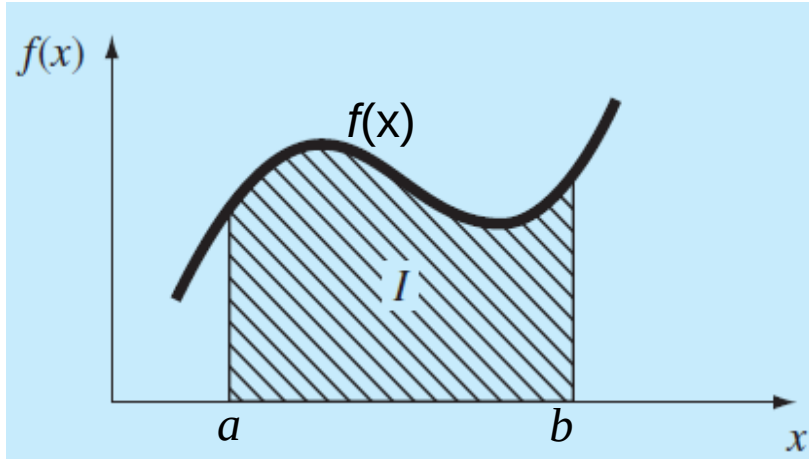


- Fit curves to data points
- Directly through the data points and determine intermediate values between data points

- Fit deterministic computer simulation data
- Well-suited for interpolating tabulated data/information

SUBJECT AREAS OF THIS COURSE

Integration



Math Description

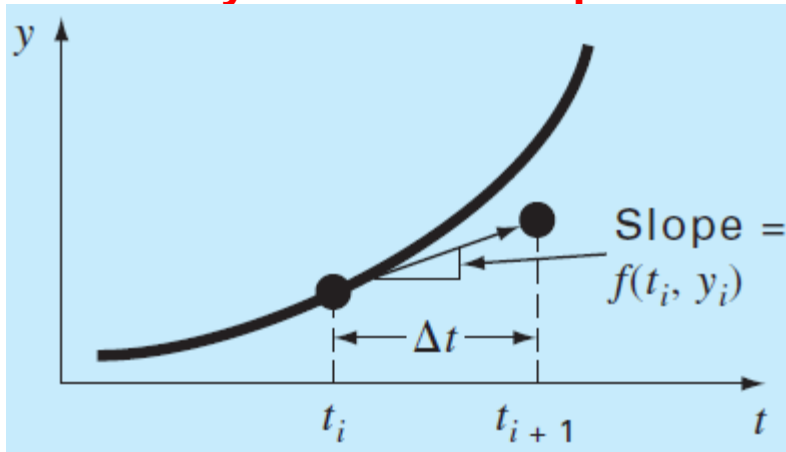
$$I = \int_a^b f(x) dx$$

Find the area under the curve

Application Examples

- Determination of centroids; total heat/flow flux through an area ...
- Solute for differential equations

Ordinary Differential Equations



Given

$$\frac{dy}{dt} \approx \frac{\Delta y}{\Delta t} = f(t, y)$$

Solve for y as a function of t .

$$y_{i+1} = y_i + f(t_i, y_i) \Delta t$$

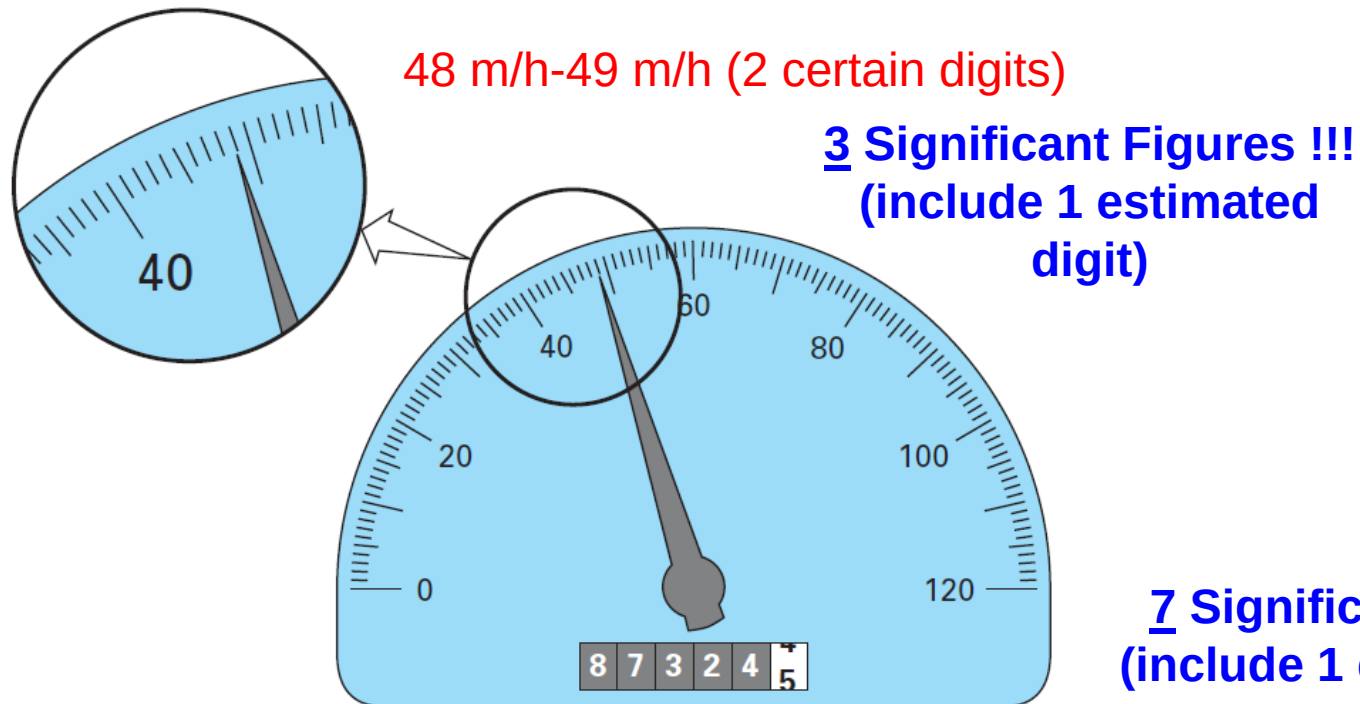
- Many physical laws are written in terms of the rate of change of a quantity
- Transient thermal, fluid, and structure problems in 1D

APPROXIMATION AND ERROR

- Numerical methods offer estimates to real solutions
 - Error is not avoidable
- Numerical errors can be classified into two categories
 - Round-off error: computers can represent only quantities with a finite number of digits
 - $\pi = 3.14159265358973 \dots$
 - Truncation error: discrepancy introduced by the fact that numerical methods employ approximations to represent exact mathematical operations and quantities
 - $\sin(x) \approx x - x^3/6$ for small x
 - $$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!}$$
- Goal of Error Analysis:
 - Identification, Quantification, and Minimization of the errors

SIGNIFICANT FIGURES: APPROXIMATE REPRESENTATION OF NUMBERS

- The significant figures/digits of a number are those that can be used with confidence
- They correspond to the number of certain digits plus one estimated digit
- It is conventional to set the estimated digit at one-half of the smallest scale division on the measurement device



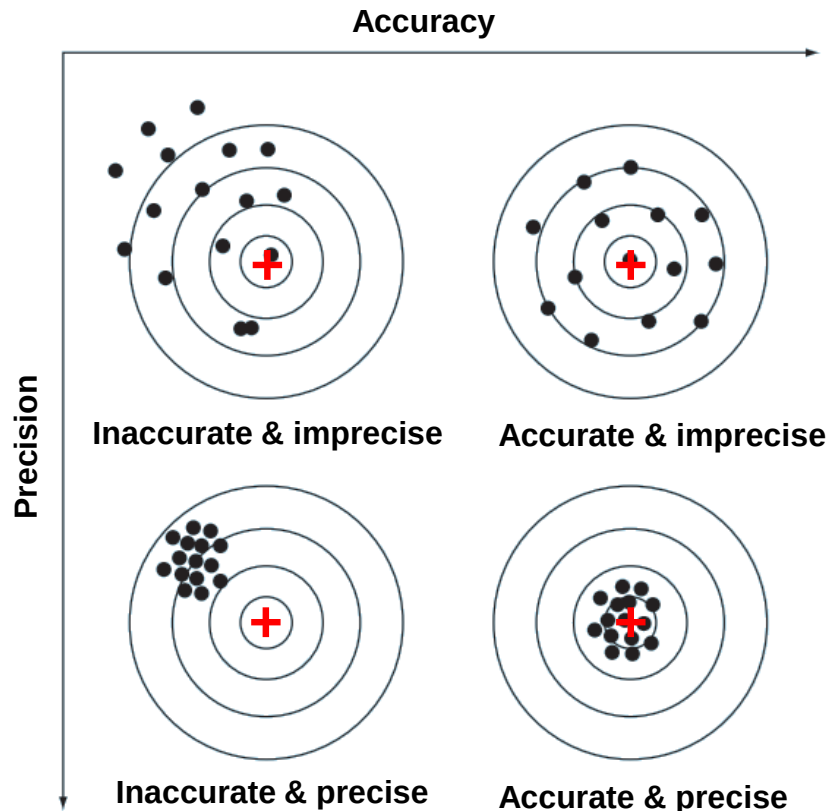
87,324.4 m-87,234.5 m (6 certain digits)

7 Significant Figures !!!
(include 1 estimated digit)

- Zeros are not always significant figures as they may be necessary to just locate a decimal point
- What is the number of the Significant figures for the following sets?
- 0.00112; 0.0112
- 10,000
- 1.0E+4, 1.00E+4

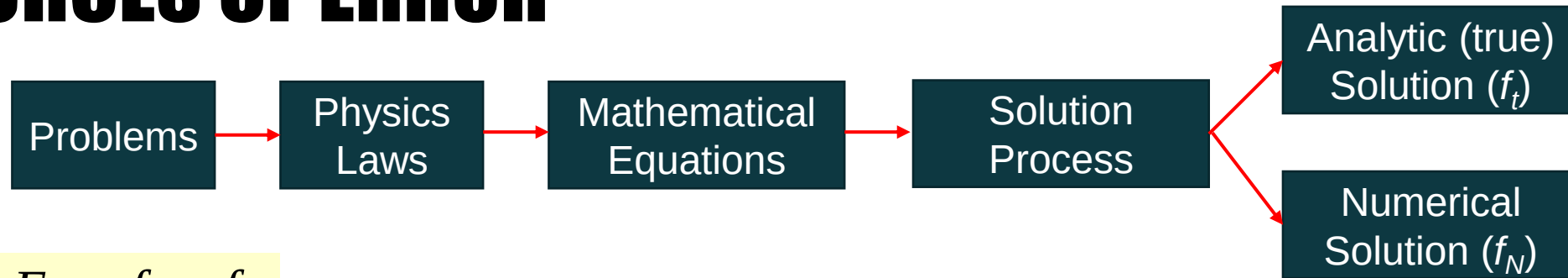
CHARACTERIZATION OF ERROR

- Accuracy: How closely a data point (computed or measured value) agrees with the **true value**
 - Inaccuracy/bias



- Precision: How closely individual data points (computed or measured values) agree with **each other**
 - Imprecision/uncertainty
- Error: including both inaccuracy (bias) and imprecision

SOURCES OF ERROR



$$\text{Error: } E_t = f_t - f_N$$

True value Numerical value

$$\text{Error: } E_t = E_{trunc} + E_{round-off} \quad (\text{True error})$$

- **Round-off error** $E_{round-off}$: computers can represent only quantities with a finite number of digits
- **Truncation error** E_{trunc} : discrepancy introduced by the fact that numerical methods employ approximations to represent exact mathematical operations and quantities

$$\text{True Fractional Relative Error: } \varepsilon_t = \left| \frac{f_t - f_N}{f_t} \right| \times 100\%$$

Relative to True value

AN EXAMPLE OF CALCULATION OF ERRORS

Length	f_t	f_N	E_t	ϵ_t
Rivet	10 cm	9 cm	1 cm	$\epsilon_t = \frac{1}{10}100\% = 10\%$
Bridge	10,000 cm	9,999 cm	1 cm	$\epsilon_t = \frac{1}{10,000}100\% = 0.01\%$

$$\text{True Fractional Relative Error: } \epsilon_t = \left| \frac{f_t - f_N}{f_t} \right| \times 100\%$$

Same absolute error can yield significantly different relative error!

APPROXIMATE ERRORS

- Issue with true error ε_t : we will not know true value/answer f_t *a priori*
- Approximate error
 - Normalize the error using the best available estimate of the true value

$$\text{Approx. Relative Error: } \varepsilon_a = \left| \frac{f_a - f_N}{f_a} \right| \times 100\%$$

Relative to estimate of true value

- Without knowledge/estimate of true value, we evaluate the difference between successive iterations (iterative approach)
 - (we hope) Approximation will become better and better, and ultimately approaches true values

"k" current approximation "k" previous approximation

$$\text{Approx. Relative Error: } \varepsilon_a = \left| \frac{f_a^k - f_a^{k-1}}{f_a^k} \right| \times 100\%$$

number of significant figures

Pre-specified Percent Tolerance

$$\varepsilon_a < \varepsilon_s$$

$$\varepsilon_s = (0.5 \times 10^{2-n})\%$$

Engineering
the result is correct to at least n significant figures

AN EXAMPLE OF APPROXIMATE ERROR

Approximate the exponential function @ $x = 0.5$ using the Maclaurin Series expansion

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!}$$

Such that the absolute value of the approximate error estimate ε_a falls below a pre-specified error criterion ε_s conforming to three significant figures ($n = 3$)

$$\text{Approx. Relative Error: } \varepsilon_a = \left| \frac{f_a^k - f_a^{k-1}}{f_a^k} \right| \times 100\%$$

Prespecified Percent Tolerance

$$\varepsilon_s = (0.5 \times 10^{2-n})\%$$

AN EXAMPLE OF APPROXIMATE ERROR

Approximate the exponential function @ $x = 0.5$ using the Maclaurin Series expansion

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!}$$

Such that the absolute value of the approximate error estimate ε_a falls below a pre-specified error criterion ε_s conforming to three significant figures ($n = 3$)

Solution: True value: $e^{0.5} = 1.648721 \dots$

Step 0: determine error criterion corresponding to three significant figures

$$\varepsilon_s = (0.5 \times 10^{2-3})\% = 0.05\%$$

Step 1: A single term $e^x = 1$

$$\varepsilon_t = \frac{1.648721 - 1}{1.648721} \times 100\% = 39.3\%$$

$$\varepsilon_a = \text{N/A}$$

Step 2: Two terms $e^x = 1 + x \Rightarrow e^{0.5} = 1 + 0.5 = 1.5$ for $x = 0.5$

$$\varepsilon_t = \frac{1.648721 - 1.5}{1.648721} \times 100\% = 9.02\%$$

$$\varepsilon_a = \frac{1.5 - 1}{1.5} \times 100\% = 33.3\%$$

because $\varepsilon_a > \varepsilon_s$, the process continues ...

AN EXAMPLE OF APPROXIMATE ERROR

Number of Terms	Numerical Approximation	ε_t (%)	ε_a (%)
1	1	39.3	
2	1.5	9.02	33.3
3	1.625	1.44	7.69
4	1.645833333	0.175	1.27
5	1.648437500	0.0172	0.158
6	1.648 <u>6</u> 97917	0.00142	0.0158

True value: 1.648721 ...

Accurate up to 5 significant figures !!!

$\varepsilon_s = (0.5 \times 10^{2-n})\%$ is conservative for this case